Factory-to-Customer Shipping Route Efficiency

Analysis for Nassau Candy Distributor

A Data Science Research Report

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Domain: Data Science — Logistics & Supply Chain Analytics

Live Dashboard: https://route-efficiency-tracker.streamlit.app/

Source Code: https://github.com/manyatri/nassau-candy-shipping-analysis

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Abstract

Nassau Candy Distributor ships chocolate, sugar, and specialty confectionery items from five factories out to customers across the United States and Canada. The company has plenty of order and shipment records sitting in its systems, but until now nobody had turned that data into route-level visibility, so logistics decisions were being made reactively rather than being backed by evidence. This report works through a factory-to-customer route efficiency analysis built on 10,194 order records, covering 196 unique factory-state routes, five ship modes, and four sales regions.

While validating the data, we ran into a problem: the raw Ship Date field was corrupted. Every shipment came out showing a lead time of somewhere between 900 and 1,640 days, and that figure had no relationship at all to the Ship Mode that was recorded. Rather than abandon the shipping-efficiency goal altogether, a Ship-Mode-conditioned Lead Time was simulated using realistic day ranges for each mode (Same Day, First Class, Second Class, Standard Class). This substitution is disclosed openly throughout the report — it is a data-quality workaround, not an authentic measurement.

Using that simulated Lead Time, routes were ranked and benchmarked, ship modes were compared, and regional bottlenecks were investigated. Reliable routes (those with 10 or more shipments) cluster tightly, averaging between 3.13 and 5.00 days, and the ship-mode results follow the pattern you would expect — Same Day comes in fastest at 0.50 days, Standard Class slowest at 5.00. The biggest finding, though, is not about geography at all — it is about factory concentration. Two of the five factories, Lot's O' Nuts and Wicked Choccy's, are responsible for 96.5% of all shipments between them, which leaves the distributor heavily dependent on just two production sites. The complete analysis is also delivered through an interactive Streamlit dashboard, with route, ship-mode, regional, and map-based views.

Keywords

Shipping Route Efficiency; Supply Chain Analytics; Logistics KPI Benchmarking; Lead Time Analysis; Geographic Bottleneck Detection; Exploratory Data Analysis; Streamlit Dashboard; Data Quality Validation.

1. Introduction

National distributors have to move goods from a handful of factories out to a customer base scattered across the map, and how fast and consistently that happens — the shipping lead time — has a real effect on customer satisfaction, operating costs, and how well the business can scale. When a distributor cannot tell which factory-to-customer routes are running smoothly and which are not, decisions about carriers, factory allocation, and regional staffing end up being made on intuition rather than on evidence.

Nassau Candy Distributor is a good example of this gap. It runs five factories, each producing a different slice of its confectionery lineup, and ships to customers spread across 59 U.S. states/provinces and two countries. The order-level data already contains everything needed to measure route performance — order and ship dates, ship mode, customer location, product identity — but none of it had ever been turned into anything operational.

This project builds that pipeline from the ground up: cleaning and validating the raw order data, engineering a route-level Lead Time metric, benchmarking routes and ship modes, flagging geographic bottlenecks, and packaging the results into a live, interactive dashboard that Nassau Candy's team can use without needing any data science background.

2. Literature Review

Routing goods efficiently from a small number of depots out to a wide set of destinations is a problem operations researchers have been chipping away at for decades. Dantzig and Ramser [1] were the first to formalize it, back in 1959, as the Truck Dispatching Problem — framing route assignment as a linear-programming exercise aimed at minimizing total distance while still meeting demand everywhere. Laporte [2] later traced fifty years of research that followed, pointing out that route-level analysis is still foundational to distribution management even though the solution methods themselves have gotten far more sophisticated.

Within supply chain management specifically, Gunasekaran and Kobu's widely cited review of performance measures [3] argues that logistics KPIs only really earn their keep when they are tied to a structured measurement framework, rather than tracked in an ad hoc way — a principle this project follows by defining Lead Time, Route Volume, and Delay Frequency as explicit, reproducible KPIs instead of one-off numbers pulled together for a report. The SCOR framework from the Association for Supply Chain Management [4] takes this further, organizing supply chain performance into standardized reliability, responsiveness, and cost dimensions that make it possible to compare routes and modes on a common basis; the reliability-focused metrics used in Section 5 (average lead time, delay frequency) line up with SCOR's Delivery Performance and Order Fulfillment Cycle Time dimensions.

Standard textbooks in the field, including Chopra and Meindl [5] and Ballou [6], stress that choosing a shipping mode is fundamentally a cost-versus-responsiveness trade-off — faster modes cut down lead time but cost more per shipment — and that is exactly the framework applied in Section 6.3 when comparing Nassau Candy's four ship modes. Taken together, this literature supports two choices made in this project: first, treating route-level lead time as the core, decomposable KPI for efficiency benchmarking, in line with SCOR's reliability dimension [4]; and second, evaluating ship-mode performance as a genuine cost-speed trade-off rather than simply ranking modes by speed, consistent with [5] and [6].

3. Problem Statement

Right now, Nassau Candy Distributor does not have a clear picture of:

- Which factory-to-customer routes run consistently well, and which ones keep running into delays.
- How shipping performance changes across regions, states, and ship modes.
- Where the geographic bottlenecks actually are, and whether the busiest areas are also the slowest ones.
- How reliant the distribution network is on any single factory.

Without that visibility, any attempt to optimize logistics stays reactive instead of evidence-based. This report works through each of these gaps directly and presents the findings both as a written analysis and as a live dashboard.

4. Dataset Description

The analysis is built on Nassau Candy's own order-level shipment records: 10,194 orders placed between January 2024 and December 2025, spanning 59 U.S. states/provinces across two countries (9,994 United States orders and 200 Canada orders). None of the eighteen source fields had missing values.

Field	Description
Row ID / Order ID	Unique row and order identifiers
Order Date / Ship Date	Date the order was placed / shipped
Ship Mode	Same Day, First Class, Second Class, Standard Class
Customer ID, City, State/Province, Postal Code	Customer identity and geography

Field	Description
Division, Product ID, Product Name	Product classification and identity
Region	Sales region — Pacific, Atlantic, Interior, Gulf
Sales, Units, Gross Profit, Cost	Transaction value and profitability

Table 1: Source dataset fields used in the analysis.

Two reference tables came along with the order data and turned out to be central to how routes are defined in Section 5.2: a factory location table (five factories, each with geographic coordinates) and a product-to-factory mapping table linking each of the fifteen catalog products, across three divisions (Chocolate, Sugar, Other), to the single factory that produces it.

Chocolate dominates the catalog: 9,844 of the 10,194 orders (96.6%) are for Chocolate-division products, with Sugar and Other combined accounting for the remaining 3.4%. Altogether, the dataset represents $141,783.63 in total sales, $93,442.80 in gross profit (a 65.9% margin), and 38,654 units shipped.

5. Methodology

5.1 Data Cleaning & Validation

Dates were parsed out of their raw DD-MM-YYYY string format into proper datetime values, and geographic fields (state and province names) were standardized so they would group consistently. During validation, a data-quality issue turned up that could not be ignored: Order Date values sit within 2024–2025, while Ship Date values land somewhere in 2026–2030 — a systematic offset that produced an apparent shipping lead time of 904 to 1,642 days per order, with no meaningful tie to the stated Ship Mode (even orders marked 'Same Day' showed multi-year apparent lead times). That is not a plausible shipping duration for a confectionery distributor by any stretch, so it was treated as a data-generation defect rather than a genuine finding.

Instead of computing Lead Time directly from these corrupted fields — or dropping the shipping-efficiency objective altogether — a Ship-Mode-conditioned Lead Time was simulated: each order was assigned a lead time drawn from a range appropriate to its stated Ship Mode (Same Day: 0–1 days; First Class: 1–3 days; Second Class: 2–5 days; Standard Class: 3–7 days), based on realistic U.S. parcel-carrier service windows. This substitution is disclosed here plainly and should be read as a modeled approximation used to demonstrate the analytical pipeline, not as an authentic measurement of Nassau Candy's actual shipping performance. A production deployment of this pipeline would need corrected source timestamps before the Lead Time KPI could be used operationally.

5.2 Feature Engineering

Three engineered fields drive the rest of the analysis:

- Lead Time (Days) — the Ship-Mode-conditioned simulated duration described in Section 5.1.
- Factory — worked out by mapping each order's Product Name to its producing factory using the supplied product-factory reference table.
- Route — the combination of Factory and customer State/Province (e.g., "Lot's O' Nuts → Ohio"), which is the basic unit this whole report analyzes.

5.3 Route Definition & Aggregation

Each of the 10,194 orders was grouped by its Route, producing 196 unique factory-to-state combinations. For every route, three summary statistics were computed: Total Shipments (order volume), Average Lead Time, and Lead Time standard deviation (variability). Routes with very few orders produce averages that are not statistically stable — a single order that happens to finish in 0 days could make a route look like the "fastest" purely by chance — so a reliability threshold of at least 10 shipments was applied before ranking anything, leaving 96 of the 196 routes eligible for the leaderboard in Section 6.2.

5.4 Efficiency Benchmarking

Reliable routes were ranked from fastest to slowest by average Lead Time to pull out the ten most and least efficient (Section 6.2), and performance was separately compared across the four Ship Modes (Section 6.3) to look at the trade-off between expedited and standard shipping.

5.5 Geographic Bottleneck Analysis

Shipments were aggregated by sales Region to look for areas that combine high volume with poor performance — that is the working definition of a bottleneck used in this report (Section 6.4). Factory-level order share was examined separately, since that is a question of structural rather than geographic concentration risk (Section 6.5).

5.6 Streamlit Dashboard Development

The complete analysis was deployed as an interactive Streamlit web application with Plotly visualizations, giving Nassau Candy's team live, filterable access to the same route, ship-mode, regional, and map-based views presented in this report, without requiring any code or data science background. Dashboard filters include Ship Mode, Region, and a configurable delay threshold; details are given in Section 6.6.

6. Results & Discussion

6.1 Exploratory Data Analysis

- The dataset covers 10,194 orders across two countries, 59 states/provinces, and 196 unique factory-to-state routes.

- Chocolate-division products dominate the catalog, accounting for 96.6% of all orders; Sugar and Other divisions are minor by comparison.
- Standard Class is the default ship mode for 60.0% of all orders, with expedited modes (Same Day, First Class, Second Class) together covering the remaining 40.0%.
- Total sales across the analyzed period reached $141,783.63, with a gross profit margin of 65.9% — a healthy margin typical of confectionery distribution.

6.2 Route Efficiency Leaderboard

Among the 96 routes meeting the ≥10-shipment reliability threshold, average Lead Time ranges narrowly from 3.13 to 5.00 days — a comparatively tight spread, which suggests that under the simulated Lead Time model, no single reliable route is a severe outlier. Worth noting: Lot's O' Nuts → Ohio is both the highest-volume reliable route (262 shipments) and one of the fastest (3.46 days), which shows that high demand does not, in this dataset, necessarily line up with slower service.

Route	Shipments	Avg. Lead Time (days)
Lot's O' Nuts → Rhode Island	30	3.13
Lot's O' Nuts → Quebec	24	3.25
Wicked Choccy's → Louisiana	15	3.27
Wicked Choccy's → Connecticut	31	3.29
Secret Factory → Illinois	13	3.31
Wicked Choccy's → Rhode Island	21	3.33
Wicked Choccy's → Nebraska	14	3.36
Wicked Choccy's → Missouri	29	3.41
Wicked Choccy's → New Hampshire	11	3.45
Lot's O' Nuts → Ohio	262	3.46

Table 2: Top 10 fastest reliable routes (≥10 shipments).

Route	Shipments	Avg. Lead Time (days)
Lot's O' Nuts → Minnesota	43	4.51
Lot's O' Nuts → Iowa	16	4.56
Wicked Choccy's → Wisconsin	47	4.57
Wicked Choccy's → Tennessee	67	4.58
Lot's O' Nuts → Alabama	34	4.59

Route	Shipments	Avg. Lead Time (days)
Wicked Choccy's → Mississippi	22	4.64
Lot's O' Nuts → New Mexico	18	4.67
Wicked Choccy's → Indiana	54	4.70
Wicked Choccy's → Ontario	30	4.87
Lot's O' Nuts → British Columbia	18	5.00

Table 3: Bottom 10 slowest reliable routes (≥10 shipments).

6.3 Ship Mode Comparison

Average Lead Time climbs steadily from Same Day through Standard Class, exactly as expected given the way the simulation was built, confirming that the modeled data still preserves a logically consistent service-tier hierarchy for the purpose of illustrating the analytical pipeline.

Ship Mode	Shipments	Share	Avg. Lead Time (days)
Same Day	547	5.4%	0.50
First Class	1,548	15.2%	2.02
Second Class	1,979	19.4%	3.48
Standard Class	6,120	60.0%	5.00

Table 4: Shipment volume and average lead time by Ship Mode.

Standard Class carries 60% of all volume despite being the slowest tier, which fits the classic cost-responsiveness trade-off Chopra and Meindl describe [5]: most customers, or most order types, default to the lower-cost, slower option rather than pay extra for expedited service.

6.4 Geographic Bottleneck Analysis

At the regional level, Pacific carries the highest shipment volume (3,253 orders, 31.9% of total) while also recording the lowest average Lead Time (3.95 days) — basically the opposite of what a bottleneck would look like. Gulf shows the slowest average Lead Time (4.11 days) but also the lowest volume (1,620 orders), so it does not fit the bottleneck pattern either.

Region	Shipments	Avg. Lead Time (days)	Total Sales
Pacific	3,253	3.95	$46,301.53
Atlantic	2,986	3.96	$41,197.24
Interior	2,335	4.08	$32,037.60
Gulf	1,620	4.11	$22,247.26

Table 5: Regional shipment volume, lead time, and sales.

Using the volume-plus-delay definition of a bottleneck adopted in this report, no region really qualifies as a serious operational bottleneck — regional lead times differ by only 0.16 days (3.95 to 4.11) across the full range, a gap too small to be operationally meaningful given the coarse day-level granularity of the simulated Lead Time.

6.5 Factory Dependency: The Central Finding

The most consequential pattern in the data is not geographic — it is structural. Two of Nassau Candy's five factories, Lot's O' Nuts and Wicked Choccy's, together produce 96.5% of all shipped orders, while the remaining three factories (Secret Factory, The Other Factory, Sugar Shack) combined account for just 3.5%.

Factory	Shipments	Share of Total
Lot's O' Nuts	5,692	55.8%
Wicked Choccy's	4,152	40.7%
Secret Factory	217	2.1%
The Other Factory	100	1.0%
Sugar Shack	33	0.3%

Table 6: Order volume by factory.

What this means practically is that any disruption at either of the two dominant factories — equipment failure, a labor action, weather, a regional disaster — would hit the large majority of Nassau Candy's order fulfillment capacity all at once. This risk matters more to overall shipping reliability than any of the route- or region-level lead-time differences observed in Sections 6.2–6.4, and it is the report's primary operational recommendation (Section 7).

6.6 Streamlit Dashboard

The full analysis is deployed as a multi-tab interactive Streamlit dashboard: a Route Leaderboard tab with an adjustable minimum-shipment reliability slider; a Ship Mode Comparison tab with a bar chart of average lead time by mode; a Regional Performance tab with a volume-versus-lead-time scatter plot for bottleneck screening; and a Map tab with a state-level choropleth of average lead time. Sidebar filters for Ship Mode, Region, and delay threshold let Nassau Candy's team interactively reproduce every table and chart in this report, and drill into specific segments not covered here.

7. Recommendations for Nassau Candy

- Treat factory concentration, not geography, as the top operational risk: develop contingency or backup-production plans for Lot's O' Nuts and Wicked Choccy's given their combined 96.5% share of order volume.
- Investigate onboarding barriers at the three low-volume factories (Secret Factory, The Other Factory, Sugar Shack) — expanding their share of production would reduce single-point-of-failure exposure without requiring new capital facilities.
- Continue to promote Standard Class as the default tier given its cost-effectiveness, but monitor the delay-frequency KPI for high-value or time-sensitive orders where the 5.00-day average may not meet customer expectations.
- Before using shipping Lead Time as an operational KPI, correct the underlying Ship Date data-capture defect identified in Section 5.1; the simulated figures in this report are suitable for demonstrating the analytical pipeline but not for live decision-making.
- Adopt the Streamlit dashboard as a standing tool for the logistics team so that route- and factory-level visibility persists beyond this one-time report.

8. Conclusion & Future Scope

This project sets up a repeatable, data-driven way of understanding shipping route efficiency at Nassau Candy Distributor. Cleaning and validating the raw order data surfaced a critical data-quality defect in the Ship Date field, which was addressed transparently through a disclosed simulation approach rather than quietly worked around — that transparency is itself a central finding of this report. Applying route-, mode-, and region-level aggregation to the resulting Lead Time metric shows a distribution network that is geographically well-balanced but structurally concentrated in two of five factories, a risk that outweighs any route-level delay pattern observed in the data.

Future work should prioritize three directions: first, correcting the Ship Date capture defect at the source so that Lead Time can be computed directly from authentic timestamps rather than simulated ranges; second, extending the reliability threshold analysis with formal statistical testing (confidence intervals per route, for example) rather than a fixed minimum-shipment cutoff; and third, layering a vehicle-routing-style optimization model [1], [2] on top of the current descriptive benchmarking to recommend concrete route or carrier reallocations rather than only reporting historical performance.

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